

Determining the angle between 2 lines

①

ex: Determine the angle between the 2 lines below. They intersect at the point $(-1, 10, 3)$

$$l_1: x = 3 + 2k$$

$$y = 4 - 3k$$

$$z = 5 + k$$

$$l_2: x = -7 + 6t$$

$$y = 5 + 5t$$

$$z = -1 + 4t$$

Solution:

$$\vec{d}_1 = [2, -3, 1] ; \vec{d}_2 = [6, 5, 4]$$

$$\cos(\theta) = \frac{\vec{d}_1 \cdot \vec{d}_2}{|\vec{d}_1| |\vec{d}_2|}$$

$$\cos(\theta) = \frac{[2, -3, 1] \cdot [6, 5, 4]}{\sqrt{(2)^2 + (-3)^2 + (1)^2} \cdot \sqrt{(6)^2 + (5)^2 + (4)^2}}$$

$$\cos(\theta) = \frac{1}{\sqrt{14} \cdot \sqrt{77}}$$

$$\theta \doteq 88.25^\circ \text{ (acute angle)}$$

Determining the angle between 2 planes

(2)

ex: Determine the angle between the following 2 planes.

$$\pi_1: x - y - z - 1 = 0 ; \pi_2: 2x + 3y - z + 4 = 0$$

Solution:

$$\vec{n}_1 = [1, -1, -1] ; \vec{n}_2 = [2, 3, -1]$$

$$\cos(\theta) = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|}$$

$$\cos(\theta) = \frac{[1, -1, -1] \cdot [2, 3, -1]}{\sqrt{(1)^2 + (-1)^2 + (-1)^2} \cdot \sqrt{(2)^2 + (3)^2 + (-1)^2}}$$

$$\cos(\theta) = \frac{0}{\sqrt{3} \cdot \sqrt{14}}$$

$$\theta = 90^\circ$$

Intersection of 2 planes

①

Rules:

① comparing normals

Ⓐ If the normals are scalar multiples, then the planes are parallel.

i) If the "D-value" is also the same scalar multiple, then the planes are coincident.

ii) If the "D-value" is not the same scalar multiple the planes are distinct.

Ⓑ If the normals are not scalar multiples, then there is an intersection at a line.

note: * Let $z = t$ and solve x & y in terms of t .

(2)

Example 1: Determine the intersection, if any,
of the planes

$$\pi_1: 3x + 2y + 5z - 4 = 0$$

$$\pi_2: 9x + 6y + 15z - 12 = 0$$

Solution:

$$\vec{n}_1 = [3, 2, 5]; \quad \vec{n}_2 = [9, 6, 15]$$

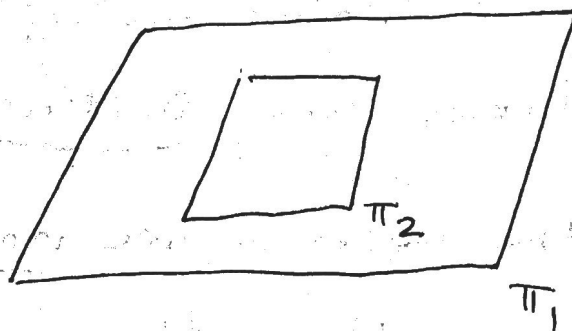
$$\therefore \vec{n}_2 = 3\vec{n}_1 \Rightarrow \pi_1 \parallel \pi_2$$

$$D_1 = -4$$

$$D_2 = -12$$

$$\Rightarrow D_2 = 3D_1 \Rightarrow \begin{matrix} \uparrow \\ \text{same} \end{matrix}$$

$\Rightarrow$ planes are
coincident.



Example 2: Determine the intersection, if any,
of the planes

$$\pi_1: x + 2y + 3z - 6 = 0$$

$$\pi_2: 4x + 8y + 12z - 25 = 0$$

Solution:

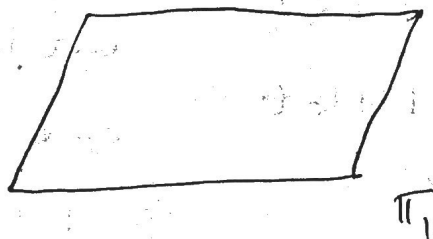
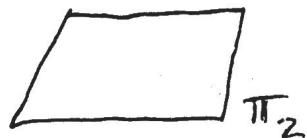
$$\vec{n}_1 = [1, 2, 3] ; \vec{n}_2 = [4, 8, 12]$$

$$\therefore \vec{n}_2 = 4\vec{n}_1 \Rightarrow \pi_1 \parallel \pi_2$$

$$D_1 = -6$$

$$D_2 = -25 \Rightarrow D_2 \neq 4D_1 \Rightarrow \text{not same scalar multiple}$$

$\therefore$ planes are parallel & distinct



Example 3: Find the parametric form of the line of intersection of the planes.

$$\pi_1: 4x - 5y - 2z - 1 = 0$$

$$\pi_2: x - y - 2z = 0$$

Solution:

$$\vec{n}_1 = [4, -5, -2] ; \vec{n}_2 = [1, -1, -2]$$

$$\vec{n}_1 \neq k\vec{n}_2, k \in \mathbb{R} \Rightarrow \text{not parallel}$$

By Elimination (Eliminate x)

$$\textcircled{1} - 4 \times \textcircled{2}$$

$$\begin{array}{r} 4x - 5y - 2z - 1 = 0 \\ -4(x - y - 2z = 0) \\ \hline -y + 6z - 1 = 0 \end{array}$$

$$\text{let } z = t$$

$$\therefore x = -1 + 8t$$

$$y = -1 + 6t$$

$$z = t$$

By Elimination (Eliminate y)

$$\textcircled{1} - 5 \times \textcircled{2}$$

$$\begin{array}{r} 4x - 5y - 2z - 1 = 0 \\ -5(x - y - 2z = 0) \\ \hline -x + 8z - 1 = 0 \end{array}$$

note: set $z = t$ since we have 2 equations and 3 unknowns.

$\Rightarrow$ 1 variable has to be arbitrarily set to t .